OpenDiscoveryTrace: Process Traces for Evaluating AI Scientist Workflows

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1. Scientific Bottleneck

AI systems now conduct scientific research at scale (Jumper et al., 2021; Merchant et al., 2023; Szymanski et al., 2023), and LLM-based agents orchestrate full research pipelines (Lu et al., 2024; Boiko et al., 2023). Benchmarks have emerged to evaluate these systems (Chen et al., 2025; Majumder et al., 2024; Starace et al., 2025; Huang et al., 2024), yet every one evaluates only *final outputs*—generated code, hypotheses, or rubric scores. The reasoning process is discarded.

This creates three problems. First, evaluating only outputs conflates reasoning quality with outcome luck. Second, governance requires complete decision traces for auditing (Kapoor & Narayanan, 2023; Hung & Yen, 2024; Tom et al., 2024). Third, improving agents requires understanding failure modes, which are trajectory properties, not output properties. OPENDISCOVERYTRACE addresses this gap with the first public dataset of structured AI scientific process traces.

2. Proposed Dataset

OPENDISCOVERYTRACE comprises **522 trajectories across 7 models**: 372 from three frontier models (GPT-5.4, Claude Opus 4.6, Gemini 3.1 Pro; 124 each, balanced across 4 domains $\times$ 3 difficulties), 120 from four open-weight models (Qwen2.5-7B, Mistral-7B-v0.3, Phi-3.5-mini, Qwen2.5-1.5B; 30 each), and 30 live-retrieval variants.

Each step records 9 fields extending ReAct (Yao et al., 2023): `phase`, `thought`, `action` (tool/input/output), `observation`, `error`, `revision_trigger`, `confidence`, `step_id`, `timestamp`. Five benchmark tasks are defined: outcome prediction, error localization, claim verification, autonomy classification, and process quality scoring. Full schema in Appendix A.

Why process traces matter. All three frontier models achieve indistinguishable success rates ($\sim$69%, p=0.997), yet Claude Opus 4.6 produces 30$\times$ more errors than GPT-5.4 (2.5 vs. 0.08; H=122, $p<10^{-4}$; Cliff's δ=0.613). These errors are qualitatively different: Claude's are 66.7% tool misuse; GPT-5.4's are 83.6% reasoning errors. Output-only benchmarks miss this entirely.

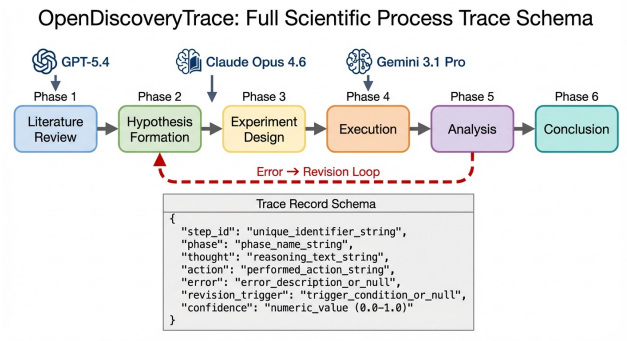

Figure 1. OPENDISCOVERYTRACE: 7 models execute tasks through a six-phase workflow. Each step is a 9-field trace. The 522 trajectories form a corpus of AI scientific process traces.

3. Acquisition Roadmap

Phase 1 (current). 522 trajectories, 7 models, 124 tasks, 4 domains. Cost: \$300 API + \$40 GPU. Schema, harness, benchmarks operational. **Phase 2 (months 1–2).** All 200 tasks $\times$ 9 models. Target: 1,800+ trajectories. **Phase 3 (months 2–3).** Human annotation on 120-trajectory sample (3 annotators, 5 axes; LLM pilot achieves α=0.63–0.82). **Phase 4 (months 3–6).** Live retrieval for literature tasks; pilot ELN integration (Dirnagl & Bhatt, 2016). **Sustainability.** Semver-versioned schema on HuggingFace with DOI. Open-source harness. Annual model refresh.

4. Metadata, Governance & Safety

License: CC-BY-4.0. **Format:** Self-contained JSON per trajectory. Datasheet per Gebru et al. (2021). **Release tiers:** (1) Full traces where provider ToS permits; (2) abstracted summaries (phase transitions, tool calls, errors) preserving $>$90% analytical utility. **Privacy:** No PII; public scientific knowledge only. **Splits:** Task-level 80/20 train/test, domain-stratified, no leakage (Kapoor & Narayanan, 2023). Evaluation follows HELM (Liang et al., 2023).

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Submitted to the AI for Science workshop (ICML 2026).

5. Acceleration Potential

OPENDISCOVERYTRACE enables: (1) process-aware evaluation that distinguishes reasoning from luck (Wang et al., 2023); (2) early failure intervention (68.1% of first errors at step 0); (3) error-type-specific model improvement; (4) process-aware reward modeling for RL; (5) trustworthy AI scientist auditing (King et al., 2009); (6) reproducible open-weight benchmarking (120 trajectories from 4 models, no API keys required).

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Table 1. Comparison of OPENDISCOVERYTRACE with existing AI science benchmarks. OPENDISCOVERYTRACE is the first benchmark to capture full process traces including error logging, revision tracking, and failure analysis alongside final output evaluation. ✓ = supported, × = not supported, ∼ = partial support.

Benchmark	Process Traces	Error Logging	Revision Tracking	Multi-Domain	Science Specific	Multi-Model
ScienceAgentBench (Chen et al., 2025)	×	×	×	✓	✓	✓
DiscoveryBench (Majumder et al., 2024)	×	×	×	✓	✓	✓
The AI Scientist (Lu et al., 2024)	×	×	×	∼	✓	✓
PaperBench (Starace et al., 2025)	×	×	×	∼	✓	✓
ResearchBench (Liu et al., 2025)	×	×	×	✓	✓	✓
SciCode (Tian et al., 2024)	×	×	×	✓	✓	✓
MLAgentBench (Huang et al., 2024)	∼	∼	×	×	∼	✓
MLE-bench (Chan et al., 2025)	×	×	×	×	×	✓
WebArena (Zhou et al., 2024)	✓	∼	×	×	×	✓
AgentBench (Liu et al., 2024)	✓	∼	×	×	×	✓
OPENDISCOVERYTRACE (ours)	✓	✓	✓	✓	✓	✓

Appendix

A. Full Trace Schema

Listing 1. Complete OPENDISCOVERYTRACE trace schema.

```
{
  "task_id": "string",
  "domain": "drug_discovery | materials_science | genomics | literature",
  "difficulty": "easy | medium | hard",
  "prompt": "string",
  "model": "gpt-5.4 | claude-opus-4-6 | gemini-3.1-pro | ...",
  "trajectory": [
    {
      "step_id": "integer", "timestamp": "ISO8601",
      "phase": "literature_review | hypothesis | experiment_design
             | execution | analysis | revision | conclusion",
      "thought": "string",
      "action": {"type": "string", "tool": "string", "input": {}, "output": {}},
      "observation": "string",
      "error": {"occurred": "boolean", "type": "string", "message": "string"},
      "revision_trigger": "string | null",
      "confidence": "float [0, 1]"
    }
  ],
  "outcome": {
    "success": "boolean", "final_claim": "string",
    "verification": {"method": "string", "result": "string", "score": "float"},
    "failure_type": "string | null", "recovery_attempted": "boolean"
  },
  "metadata": {
    "total_steps": "integer", "total_tokens": "integer",
    "total_tool_calls": "integer", "total_failures": "integer",
    "total_revisions": "integer", "wall_time_seconds": "float"
  }
}
```

B. Benchmark Comparison

C. Pilot Analysis Details

C.1. Dataset Statistics

522 trajectories: 372 frontier (124 each from GPT-5.4, Claude Opus 4.6, Gemini 3.1 Pro) + 120 open-weight (Qwen2.5-7B, Mistral-7B-v0.3, Phi-3.5-mini, Qwen2.5-1.5B; 30 each) + 30 live-retrieval. 99.7% reach conclusions. 31.2% contain errors; 34.4% contain revisions.

Table 2. Inter-model comparison (n=124 per frontier model). *** $p<0.001$, n.s. not significant.

Metric	GPT-5.4	Claude 4.6	Gemini 3.1	Sig.	δ
Steps	3.5±3.5	4.1±4.0	4.8±5.4	n.s.	0.16
Tools	2.5±3.5	2.8±3.2	3.8±5.2	n.s.	0.15
Errors	0.08±0.3	2.5±3.4	0.4±0.8	***	0.61
Revisions	0.4±0.8	1.2±2.0	1.0±2.5	***	0.24
Success	69.0%	69.8%	69.0%	n.s.	—

C.2. Failure Taxonomy

Claude: 462 errors (66.7% tool misuse, 33.1% reasoning, 0.2% hallucination). Gemini: 165 errors (72.1% reasoning, 27.9% tool). GPT-5.4: 61 errors (83.6% reasoning, 16.4% tool).

C.3. Matched-Task Within-Domain Results

Error divergence is significant in every domain ($p<0.0001$); step counts are non-significant in every domain.

C.4. Sequence-Model Baselines

LSTM: Acc=0.693±0.015, AUROC=0.422. Transformer: Acc=0.693±0.015, AUROC=0.369. Both match majority baseline.

C.5. LLM-Based Inter-Annotator Agreement

60-trajectory sample: correctness α=0.629, reasoning α=0.711, tool use α=0.717, autonomy α=0.820.

C.6. Task Examples

Drug Discovery. Easy: "Predict LogP of ibuprofen." Medium: "Find top EGFR kinase inhibitors from ChEMBL, predict ADMET." Hard: "Propose CDK4/6 inhibitor resistance mechanism in TNBC."

Materials Science. Easy: "Band gap of silicon." Medium: "Compare perovskite stability for PV." Hard: "Design HEA synthesis for ductility."

Genomics. Easy: "TP53 protein length." Medium: "KEGG enrichment for HCC genes." Hard: "BRAF V600E across cancer types."

Literature. Easy: "AlphaFold2 CASP14 accuracy." Medium: "CRISPR off-target methods review." Hard: "Gut microbiome-immunotherapy contradiction."